UNIVERSITY OF ARIZONA

Department of Aerospace & Mechanical Engineering

Project 1: Time-Developing Flow in a Parallel Plate Channel

AME 531 — Numerical Methods in Fluid Mechanics and Heat Transfer

Spring 2026

Prepared by:

[Student Name]

[Student ID]

Instructor:

Professor Farzad Mashayek

March 24, 2026

Abstract

This report presents the numerical simulation of time-developing, pressure-driven flow between two infinite parallel plates separated by a distance $2h$, a classical problem in computational fluid dynamics. The governing Navier–Stokes equation for this configuration reduces to a one-dimensional unsteady diffusion equation with a constant source term. The problem is nondimensionalized using the channel half-width and viscous diffusion scales, yielding a parameter-free formulation on the domain $-1 \leq y^* \leq 1$. An exact analytical solution is derived using separation of variables and Fourier series expansion. Three numerical methods are implemented and compared against this exact solution: the explicit Euler method, the implicit Euler method, and the fourth-order Runge–Kutta (RK4) method, all employing second-order central differencing in space. The effects of time step size Δt, spatial grid spacing Δy, and the stability parameter $r = \Delta t/(\Delta y)^2$ are systematically investigated for each method. Results confirm that the explicit Euler method is conditionally stable ($r \leq 0.5$), the implicit Euler method is unconditionally stable but accuracy-limited at large r, and the RK4 method provides superior temporal accuracy with a numerically observed stability limit near $r \approx 0.7$. Velocity profiles, volumetric flow rate, and wall shear stress are presented and show excellent agreement with the analytical solution for all stable configurations.

1 Introduction

The time development of pressure-driven flow between parallel plates is a classical problem in viscous fluid mechanics. A fluid, initially at rest between two infinite stationary plates separated by a distance $2h$, is set into motion by a suddenly imposed constant pressure gradient dp/dx. As momentum diffuses from the channel interior toward the walls, the velocity profile evolves from zero to the steady-state parabolic Poiseuille profile. Because an exact analytical solution exists for this problem, it serves as an ideal benchmark for evaluating numerical methods [2].

The governing equation is a simplified form of the incompressible Navier–Stokes equations that reduces to a one-dimensional unsteady diffusion equation with a constant source term. Following the classification in the course notes [1], this is a parabolic PDE — a marching problem that requires initial conditions and boundary conditions. The spatial derivative is discretized using the second-order central difference scheme (CDS), and three time-integration methods are employed: the explicit Euler method, which is conditionally stable with $r = \Delta t/(\Delta y)^2 \leq 0.5$ (Eq. 4.30 in [1]); the implicit Euler method, which is unconditionally stable (Section 4.3.1 in [1]); and the fourth-order Runge–Kutta method, which offers higher temporal accuracy (Section 4.1.3 in [1]).

The objectives of this project are to:

1. Nondimensionalize the governing equation and derive the exact analytical solution for velocity, flow rate, and wall shear stress.
2. Implement the explicit Euler, implicit Euler, and fourth-order Runge–Kutta methods using finite differences.
3. Compare numerical results against the exact solution and systematically investigate the effects of Δt, Δy, and $r = \Delta t/(\Delta y)^2$ on accuracy and stability.

The report is organized as follows: Section 2 presents the problem formulation and analytical solution; Section 3 describes the numerical methods; Section 4 presents results and discussion; and Section 5 provides the conclusions.

2 Problem Formulation

2.1 Problem Description

Consider the flow of an incompressible, Newtonian fluid between two infinite parallel plates separated by a distance $2h$, as illustrated in Figure 1. The fluid is initially at rest. At time $t = 0$, a constant pressure gradient dp/dx is applied in the x-direction, driving the fluid into motion. The plates are stationary (no-slip condition), and the flow is assumed to be fully developed in x and z (no variations in those directions), so the velocity field reduces to $u = u(y, t)$ only.

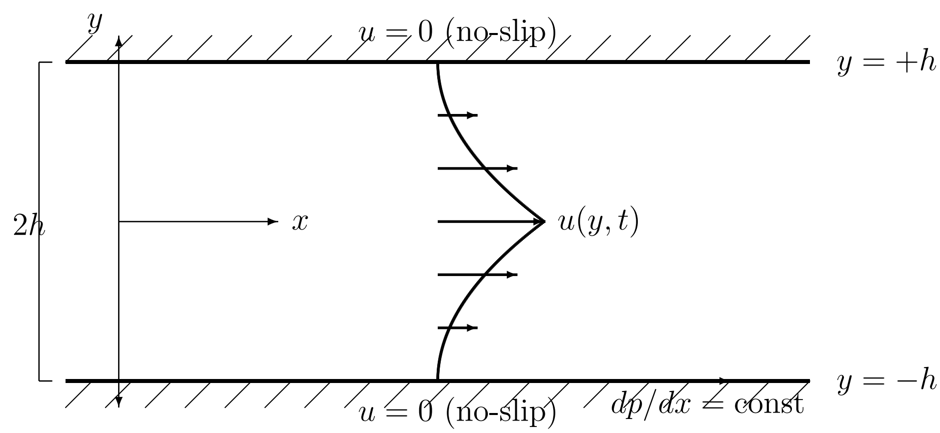

Figure 1: Schematic of time-developing flow between two parallel plates separated by distance $2h$. The fluid is driven by a constant pressure gradient dp/dx in the x-direction. Velocity arrows indicate the developing parabolic profile.

Under these assumptions, the incompressible Navier–Stokes equations simplify to:

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho}\frac{dp}{dx} + \nu \frac{\partial^2 u}{\partial y^2} \tag{1}$$

where ρ is the fluid density, $\nu = \mu/\rho$ is the kinematic viscosity, and $dp/dx < 0$ is the constant pressure gradient (negative because pressure decreases in the flow direction). The boundary and initial conditions are:

$$u(\pm h, t) = 0 \quad \text{(no-slip)}, \qquad u(y, 0) = 0 \quad \text{(fluid at rest)} \tag{2}$$

2.2 Nondimensionalization

To reduce the number of parameters and make the results universal, we introduce the following nondimensional variables:

$$y^* = \frac{y}{h}, \qquad t^* = \frac{\nu\, t}{h^2}, \qquad u^* = \frac{u}{U_{\text{ref}}} \tag{3}$$

where the reference velocity is chosen as the centerline velocity of the steady-state Poiseuille profile:

$$U_{\text{ref}} = \frac{h^2}{\mu}\left(-\frac{dp}{dx}\right) = \frac{h^2}{\rho\nu}\left(-\frac{dp}{dx}\right) \tag{4}$$

The reference time scale h^2/ν represents the characteristic viscous diffusion time across the channel half-width. With these scales, each term in Eq. (1) transforms as follows.

Left-hand side:

$$\frac{\partial u}{\partial t} = U_{\text{ref}} \cdot \frac{\nu}{h^2} \cdot \frac{\partial u^*}{\partial t^*} = \frac{1}{\rho}\left(-\frac{dp}{dx}\right)\frac{\partial u^*}{\partial t^*} \tag{5}$$

Pressure gradient term:

$$-\frac{1}{\rho}\frac{dp}{dx} = \frac{1}{\rho}\left(-\frac{dp}{dx}\right) \cdot 1 \tag{6}$$

Diffusion term:

$$\nu\frac{\partial^2 u}{\partial y^2} = \nu \cdot \frac{U_{\text{ref}}}{h^2} \cdot \frac{\partial^2 u^*}{\partial y^{*22}} = \frac{1}{\rho}\left(-\frac{dp}{dx}\right)\frac{\partial^2 u^*}{\partial y^{*22}} \tag{7}$$

Dividing through by the common factor $\frac{1}{\rho}(-dp/dx)$, which is nonzero, yields the nondimensional governing equation (dropping the asterisks for clarity):

$$\boxed{\frac{\partial u}{\partial t} = 1 + \frac{\partial^2 u}{\partial y^2}} \tag{8}$$

on the domain $-1 \leq y \leq 1$, with nondimensional boundary and initial conditions:

$$\boxed{u(-1,t) = 0, \qquad u(1,t) = 0, \qquad u(y,0) = 0} \tag{9}$$

The resulting equation is parameter-free, meaning the nondimensional solution $u(y,t)$ is universal for any fluid, channel width, or pressure gradient.

2.3 Exact Analytical Solution

The solution is decomposed into a steady-state part $u_s(y)$ and a transient part $u_t(y,t)$:

$$u(y,t) = u_s(y) + u_t(y,t) \tag{10}$$

2.3.1 Steady-State Solution

Setting $\partial u/\partial t = 0$ in Eq. (8):

$$\frac{d^2 u_s}{dy^2} = -1 \tag{11}$$

Integrating twice and applying $u_s(\pm 1) = 0$:

$$u_s(y) = \frac{1}{2}\left(1 - y^2\right) \tag{12}$$

This is the classical Poiseuille parabola with maximum $u_s(0) = 1/2$ at the centerline.

2.3.2 Transient Solution

Substituting $u = u_s + u_t$ into Eq. (8) and using Eq. (11), the transient part satisfies the homogeneous heat equation:

$$\frac{\partial u_t}{\partial t} = \frac{\partial^2 u_t}{\partial y^2}, \qquad u_t(\pm 1, t) = 0, \qquad u_t(y,0) = -u_s(y) = \frac{y^2 - 1}{2} \tag{13}$$

Applying separation of variables, $u_t(y,t) = Y(y)\,T(t)$, gives:

$$\frac{T'}{T} = \frac{Y''}{Y} = -\lambda \tag{14}$$

The spatial eigenvalue problem $Y'' + \lambda Y = 0$ with $Y(\pm 1) = 0$ yields the eigenvalues and eigenfunctions:

$$\lambda_n = \frac{(2n+1)^2 \pi^2}{4}, \qquad Y_n(y) = \cos\left(\frac{(2n+1)\pi y}{2}\right), \qquad n = 0, 1, 2, \ldots \tag{15}$$

Note that cosine functions appear because the initial condition $u_t(y,0) = (y^2 - 1)/2$ is symmetric about $y = 0$, and cosines are the symmetric eigenfunctions satisfying the homogeneous boundary conditions on $[-1, 1]$.

The temporal equation gives $T_n(t) = e^{-\lambda_n t}$. The general solution is:

$$u_t(y,t) = \sum_{n=0}^{\infty} B_n \cos\left(\frac{(2n+1)\pi y}{2}\right) e^{-(2n+1)^2\pi^2 t/4} \tag{16}$$

The coefficients B_n are determined from the initial condition using Fourier orthogonality:

$$B_n = \int_{-1}^{1} \frac{y^2-1}{2} \cos\left(\frac{(2n+1)\pi y}{2}\right) dy \Big/ \int_{-1}^{1} \cos^2\left(\frac{(2n+1)\pi y}{2}\right) dy \tag{17}$$

Evaluating via integration by parts (the denominator equals 1), the result is:

$$B_n = \frac{(-1)^{n+1}\cdot 16}{(2n+1)^3\,\pi^3} \tag{18}$$

2.3.3 Complete Solution

Combining the steady and transient parts:

$$\boxed{u(y,t) = \frac{1}{2}(1-y^2) - \frac{16}{\pi^3}\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3} \cos\left(\frac{(2n+1)\pi y}{2}\right) e^{-(2n+1)^2\pi^2 t/4}} \tag{19}$$

At $t = 0$, the series exactly cancels the parabola, giving $u = 0$ (initial condition). As $t \to \infty$, all exponentials decay to zero, leaving the steady Poiseuille profile $u_s = (1-y^2)/2$.

2.4 Analytical Flow Rate

The nondimensional volumetric flow rate per unit width is:

$$Q(t) = \int_{-1}^{1} u(y,t)\,dy \tag{20}$$

Integrating the steady part:

$$\int_{-1}^{1} \frac{1}{2}(1-y^2)\,dy = \frac{2}{3} \tag{21}$$

Integrating the transient terms (using $\int_{-1}^{1} \cos\left(\frac{(2n+1)\pi y}{2}\right) dy = \frac{4(-1)^n}{(2n+1)\pi}$):

$$\boxed{Q(t) = \frac{2}{3} - \frac{64}{\pi^4}\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4}\, e^{-(2n+1)^2\pi^2 t/4}} \tag{22}$$

The flow rate rises from $Q(0) = 0$ to the steady-state value $Q(\infty) = 2/3$.

2.5 Analytical Wall Shear Stress

The wall shear stress is defined as $\tau = -\partial u/\partial y\big|_{\text{wall}}$. Evaluating at the right wall $y = +1$:

$$\frac{\partial u}{\partial y} = -y + \frac{16}{\pi^3}\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3} \cdot \frac{(2n+1)\pi}{2} \sin\left(\frac{(2n+1)\pi y}{2}\right) e^{-(2n+1)^2\pi^2 t/4} \tag{23}$$

At $y = 1$: $\sin\left(\frac{(2n+1)\pi}{2}\right) = (-1)^n$, and $(-1)^n \cdot (-1)^n = 1$, so:

$$\left.\frac{\partial u}{\partial y}\right|_{y=1} = -1 + \frac{8}{\pi^2}\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}\, e^{-(2n+1)^2\pi^2 t/4} \tag{24}$$

Therefore:

$$\boxed{\tau(t) = -\left.\frac{\partial u}{\partial y}\right|_{y=1} = 1 - \frac{8}{\pi^2}\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}\, e^{-(2n+1)^2\pi^2 t/4}} \tag{25}$$

Verification: at $t = 0$, using $\sum_{n=0}^{\infty} 1/(2n+1)^2 = \pi^2/8$, we get $\tau(0) = 1 - 1 = 0$ (no shear when fluid is at rest). As $t \to \infty$, $\tau \to 1$ (steady-state Poiseuille wall shear). Both limits are physically consistent.

3 Numerical Formulation

3.1 Grid and Notation

The domain $-1 \le y \le 1$ is discretized into N uniform intervals, producing $N+1$ grid points:

$$y_i = -1 + i\,\Delta y, \qquad i = 0, 1, \ldots, N, \qquad \Delta y = \frac{2}{N} \tag{26}$$

Time is advanced in uniform steps Δt, so $t^n = n\,\Delta t$ for $n = 0, 1, 2, \ldots$. The numerical approximation of $u(y_i, t^n)$ is denoted u_i^n. The key stability parameter is:

$$r = \frac{\Delta t}{(\Delta y)^2} \tag{27}$$

The boundary conditions $u_0^n = u_N^n = 0$ are enforced at every time level. The initial condition is $u_i^0 = 0$ for all i.

3.2 Spatial Discretization

The second derivative in Eq. (8) is approximated at interior nodes $i = 1, 2, \ldots, N-1$ using the second-order central difference scheme (CDS), derived from the Taylor series expansion (Chapter 3 of [1], Eq. 3.46):

$$\left.\frac{\partial^2 u}{\partial y^2}\right|_{y_i} \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta y)^2} + \mathcal{O}\big((\Delta y)^2\big) \tag{28}$$

This approximation has a truncation error of order $(\Delta y)^2$, meaning the spatial discretization is second-order accurate.

3.3 Explicit Euler Method

The explicit (forward) Euler method evaluates the right-hand side at the current known time level n (Section 4.2.1 of [1]):

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = 1 + \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{(\Delta y)^2} \tag{29}$$

Rearranging to isolate u_i^{n+1}:

$$\boxed{u_i^{n+1} = r\,u_{i-1}^n + (1 - 2r)\,u_i^n + r\,u_{i+1}^n + \Delta t} \tag{30}$$

This is a direct update: each new value is computed explicitly from the known values at time level n. No system of equations needs to be solved.

Stability. Von Neumann stability analysis (Section 4.2.1 of [1], Eq. 4.30) shows that the explicit Euler method applied to the diffusion equation requires:

$$r = \frac{\Delta t}{(\Delta y)^2} \leq \frac{1}{2} \tag{31}$$

This imposes a severe constraint: halving Δy requires reducing Δt by a factor of four.

3.4 Implicit Euler Method

The implicit (backward) Euler method evaluates the spatial derivative at the unknown future time level $n+1$ (Section 4.3.1 of [1]):

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = 1 + \frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{(\Delta y)^2} \tag{32}$$

Rearranging with all $n+1$ terms on the left:

$$\boxed{-r\, u_{i-1}^{n+1} + (1+2r)\, u_i^{n+1} - r\, u_{i+1}^{n+1} = u_i^n + \Delta t} \tag{33}$$

This constitutes a tridiagonal system of $(N-1)$ equations for the interior unknowns $u_1^{n+1}, u_2^{n+1}, \ldots, u_{N-1}^{n+1}$. In matrix form:

$$\underbrace{\begin{pmatrix} 1{+}2r & -r & & \\ -r & 1{+}2r & -r & \\ & \ddots & \ddots & \ddots \\ & & -r & 1{+}2r \end{pmatrix}}_{\mathbf{M}} \begin{pmatrix} u_1^{n+1} \\ u_2^{n+1} \\ \vdots \\ u_{N-1}^{n+1} \end{pmatrix} = \begin{pmatrix} u_1^n + \Delta t \\ u_2^n + \Delta t \\ \vdots \\ u_{N-1}^n + \Delta t \end{pmatrix} \tag{34}$$

The boundary values $u_0^{n+1} = u_N^{n+1} = 0$ are already incorporated (they contribute zero to the first and last equations). This system is solved efficiently at each time step using the Thomas algorithm (tridiagonal matrix algorithm), which requires only $\mathcal{O}(N)$ operations [2].

Stability. The implicit Euler method is unconditionally stable (Section 4.3.1 of [1]): the solution remains bounded for any value of r. However, the method is first-order accurate in time, so large Δt degrades accuracy despite maintaining stability.

3.5 Fourth-Order Runge–Kutta Method

The governing equation is rewritten in the semi-discrete form:

$$\frac{d\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u}), \qquad \text{where} \quad f(u_i) = 1 + \frac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta y)^2} \tag{35}$$

for interior points $i = 1, \ldots, N-1$, with $f(u_0) = f(u_N) = 0$.

The classical four-stage Runge–Kutta method (Section 4.1.3 of [1], Eq. 4.16) advances the solution from t^n to t^{n+1} as follows:

$$\boxed{\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(\mathbf{u}^n) \\ \mathbf{k}_2 &= \mathbf{f}\big(\mathbf{u}^n + \tfrac{\Delta t}{2}\,\mathbf{k}_1\big) \\ \mathbf{k}_3 &= \mathbf{f}\big(\mathbf{u}^n + \tfrac{\Delta t}{2}\,\mathbf{k}_2\big) \\ \mathbf{k}_4 &= \mathbf{f}(\mathbf{u}^n + \Delta t\,\mathbf{k}_3) \\ \mathbf{u}^{n+1} &= \mathbf{u}^n + \frac{\Delta t}{6}\,(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \end{aligned}} \tag{36}$$

The boundary conditions $u_0 = u_N = 0$ are enforced after each intermediate stage (i.e., on the arguments of $\mathbf{f}$ at stages 2, 3, and 4) and after the final update.

Accuracy. The RK4 method is fourth-order accurate in time: the truncation error is $\mathcal{O}((\Delta t)^4)$. Combined with the second-order CDS in space, the overall scheme is $\mathcal{O}((\Delta t)^4 + (\Delta y)^2)$.

Stability. The RK4 method is explicit and therefore conditionally stable. The course notes (Section 4.1.3 of [1]) state that the method has a stability restriction but do not provide the specific limit for the diffusion equation. In this work, the stability boundary is determined numerically and found to occur near $r \approx 0.7$.

Cost. Each time step requires four evaluations of $\mathbf{f}$, compared to one for the explicit Euler and one tridiagonal solve for the implicit Euler.

3.6 Numerical Flow Rate

The volumetric flow rate $Q(t) = \int_{-1}^{1} u\,dy$ is approximated at each time step using the trapezoidal rule:

$$Q^n \approx \Delta y \left[\frac{u_0^n}{2} + \sum_{i=1}^{N-1} u_i^n + \frac{u_N^n}{2}\right] = \Delta y \sum_{i=1}^{N-1} u_i^n \tag{37}$$

where the last equality follows because $u_0^n = u_N^n = 0$. The trapezoidal rule is second-order accurate, consistent with the spatial discretization.

3.7 Numerical Wall Shear Stress

The wall shear stress $\tau = -\partial u/\partial y\big|_{y=1}$ is computed at the right wall using the second-order backward (one-sided) finite difference (Chapter 3 of [1], Eq. 3.62):

$$\left.\frac{\partial u}{\partial y}\right|_{y_N} \approx \frac{3u_N - 4u_{N-1} + u_{N-2}}{2\,\Delta y} + \mathcal{O}\big((\Delta y)^2\big) \tag{38}$$

Since $u_N = 0$, this simplifies to:

$$\boxed{\tau^n = -\frac{-4u^n_{N-1} + u^n_{N-2}}{2\,\Delta y} = \frac{4u^n_{N-1} - u^n_{N-2}}{2\,\Delta y}} \tag{39}$$

This second-order one-sided stencil is used because the wall is at the domain boundary, where a centered difference is not available.

3.8 Summary of Methods

Table 1 provides a concise comparison of the three numerical methods.

Table 1: Comparison of the three time-integration methods.

Property	**Explicit Euler**	**Implicit Euler**	**RK4**
Time accuracy	$\mathcal{O}(\Delta t)$	$\mathcal{O}(\Delta t)$	$\mathcal{O}((\Delta t)^4)$
Space accuracy	$\mathcal{O}((\Delta y)^2)$	$\mathcal{O}((\Delta y)^2)$	$\mathcal{O}((\Delta y)^2)$
Stability	$r \leq 0.5$	Unconditional	$r \lesssim 0.7$ (numerical)
System to solve	None	Tridiagonal	None
RHS evals/step	1	1	4

4 Results and Discussion

All simulations are performed with the Python programming language using NumPy for array operations. Unless otherwise stated, the baseline parameters are $N = 40$ ($\Delta y = 0.05$), $\Delta t = 0.001$ ($r = 0.4$), and $t^*_{\text{final}} = 1.0$. The exact analytical solution is evaluated with 200 Fourier terms, which provides convergence to machine precision.

4.1 Velocity Profiles

Figures 2–4 show the evolution of the velocity profile from rest ($t^* = 0$) toward the steady-state Poiseuille parabola for the three numerical methods. Each figure displays profiles at six time instants ($t^* = 0.01, 0.05, 0.1, 0.2, 0.5, 1.0$), with the exact analytical solution shown as solid lines and numerical results as circular markers.

At early times ($t^* = 0.01$), the velocity is nearly zero everywhere except very close to the channel center, reflecting the initial diffusion of momentum from the pressure-driven interior. By $t^* = 0.1$, a recognizable but flattened profile has developed. At $t^* = 1.0$, the profile closely approaches the steady parabola, although a small difference remains (the exponential decay time for the first Fourier mode is $4/\pi^2 \approx 0.405$).

For all three methods at the baseline parameters ($r = 0.4$), the numerical markers fall directly on the exact curves, indicating excellent agreement. The differences between methods are not visible at this resolution and are quantified in the error analysis below.

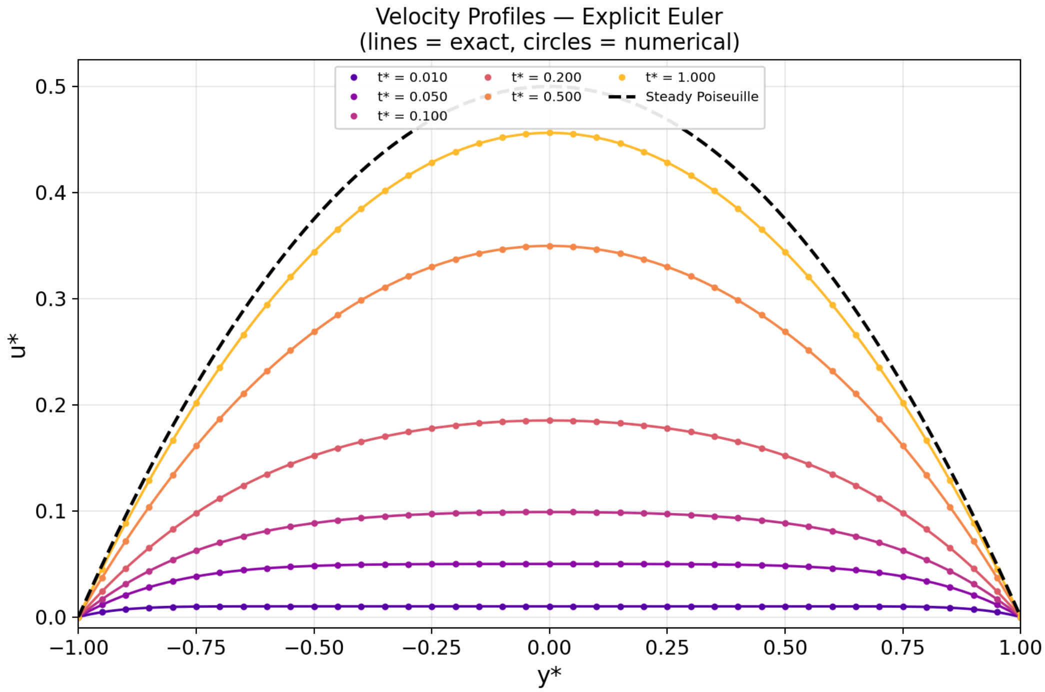

Figure 2: Velocity profiles at six time instants computed using the **explicit Euler** method ($N = 40$, $r = 0.4$).

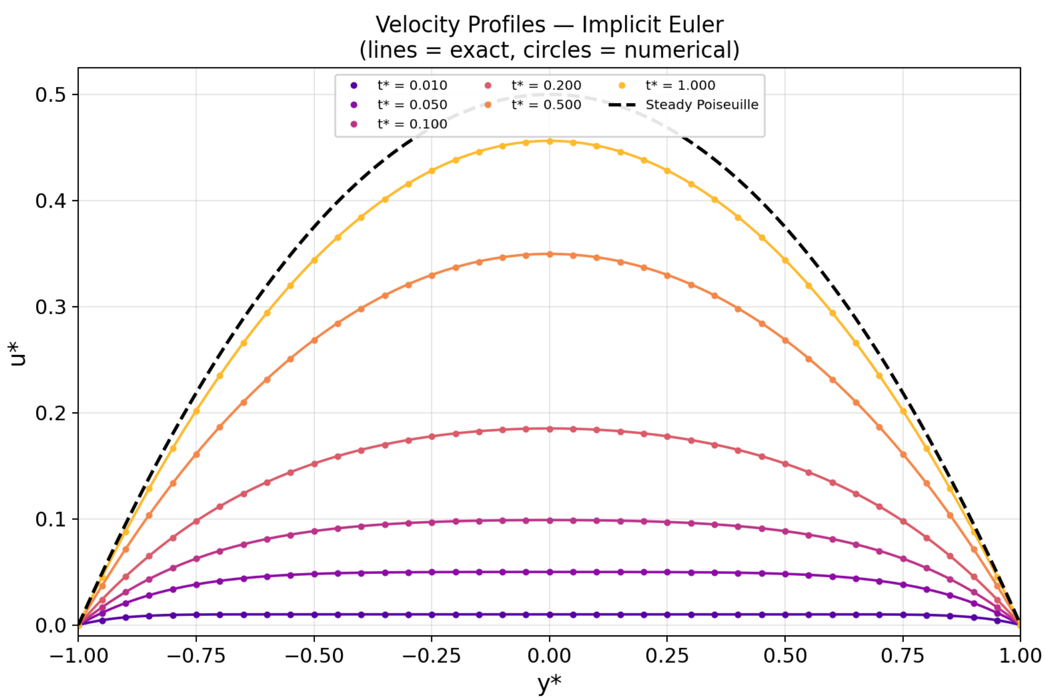

Figure 3: Velocity profiles at six time instants computed using the **implicit Euler** method ($N = 40$, $r = 0.4$).

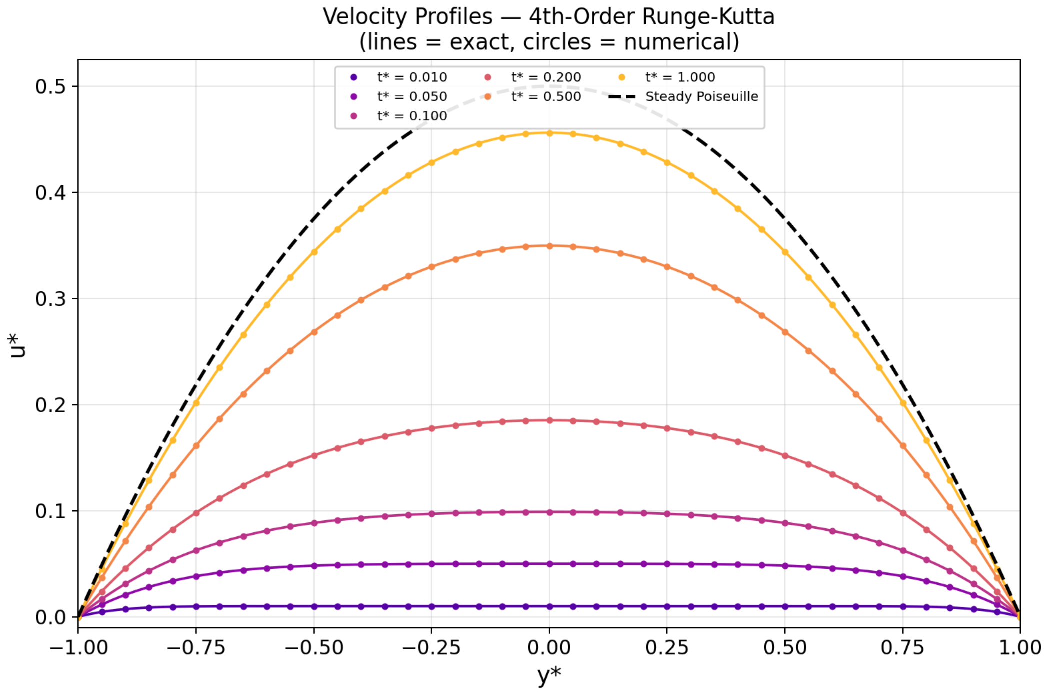

Figure 4: Velocity profiles at six time instants computed using the **fourth-order Runge–Kutta** method ($N = 40$, $r = 0.4$).

4.2 Flow Rate and Wall Shear Stress

Figure 5 presents the volumetric flow rate $Q(t^*)$ and Figure 6 presents the wall shear stress $\tau(t^*)$ evaluated at $y = +1$ (right wall). Both quantities evolve monotonically from zero at $t^* = 0$ toward their respective steady-state values of $Q_\infty = 2/3$ and $\tau_\infty = 1$.

The flow rate rises rapidly at early times as the pressure gradient accelerates the bulk fluid, then gradually levels off as viscous diffusion establishes the steady parabolic profile. The wall shear stress exhibits a similar transient, rising steeply as the near wall velocity gradient builds up. All three numerical methods produce results that are visually indistinguishable from the exact solution on these plots.

Table 2 quantifies the errors at the final time $t^* = 1.0$ for the baseline parameters. The RK4 method achieves the smallest E_∞ error, followed by the explicit Euler and then the implicit Euler, consistent with their respective temporal accuracy orders.

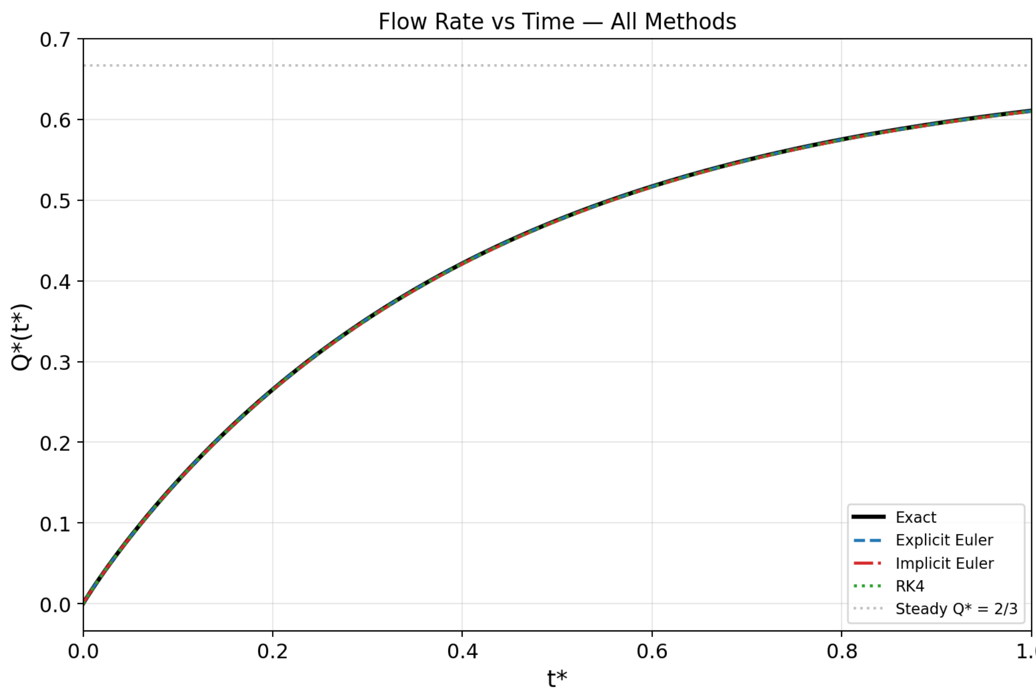

Figure 5: Volumetric flow rate $Q(t^*)$ versus nondimensional time for all three numerical methods compared with the exact solution. The dashed horizontal line indicates the steady-state value $Q_\infty = 2/3$.

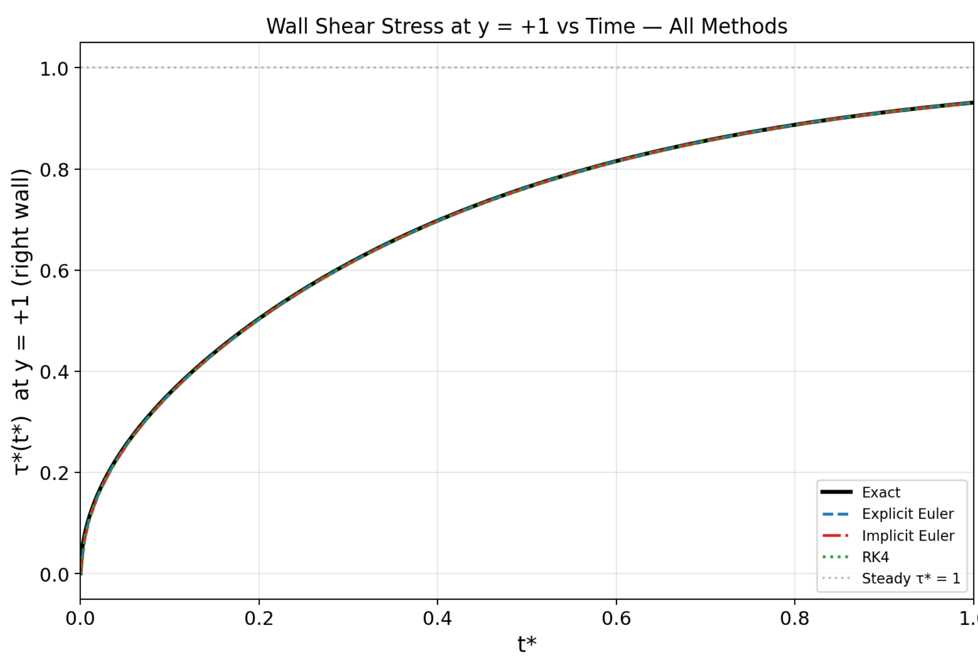

Figure 6: Wall shear stress $\tau(t^*)$ at $y = +1$ versus nondimensional time for all three methods compared with the exact solution. The dashed horizontal line indicates the steady state value $\tau_\infty = 1$.

Table 2: Error metrics at $t^* = 1.0$ for baseline parameters ($N = 40$, $\Delta t = 0.001$, $r = 0.4$).

Method	E_∞	E_{RMS}	Q_{num}	Q_{exact}	τ_{num}	τ_{exact}
Explicit Euler	7.77×10^{-5}	5.43×10^{-5}	0.61066	0.61095	0.93124	0.93126
Implicit Euler	1.89×10^{-4}	1.32×10^{-4}	0.61032	0.61095	0.93082	0.93126
RK4	5.55×10^{-5}	3.88×10^{-5}	0.61049	0.61095	0.93103	0.93126

4.3 Effect of Δt -Explicit Euler

Figure 7 shows velocity profiles at $t^* = 0.1$ computed with the explicit Euler method for four values of Δt (corresponding to $r = 0.2, 0.4, 0.5, 0.6$), with $\Delta y = 0.05$ held fixed. The three cases with $r \leq 0.5$ produce accurate, stable results that lie on top of the exact solution. The case $r = 0.6$ violates the stability condition (Eq. 31) and produces catastrophically growing oscillations, confirming the theoretical stability limit $r \leq 0.5$.

Table 3 reports the quantitative errors. The error increases approximately linearly with Δt for stable cases, consistent with the first order temporal accuracy of the explicit Euler method.

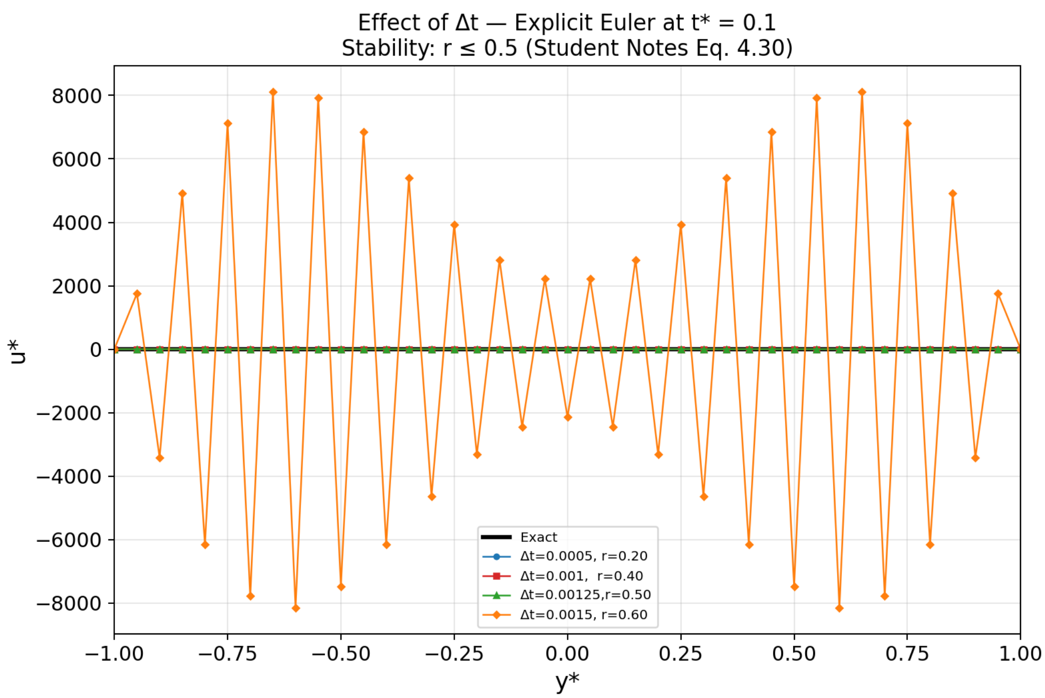

Figure 7: Effect of Δt on the explicit Euler solution at $t^* = 0.1$ ($N = 40$, $\Delta y = 0.05$). The case $r = 0.60$ exceeds the stability limit and diverges.

Table 3: Error metrics for the explicit Euler Δt study at $t^* = 0.1$.

Δt	r	E_∞	E_{RMS}	Status
0.0005	0.20	1.03×10^{-5}	7.78×10^{-6}	Stable
0.001	0.40	7.18×10^{-5}	5.43×10^{-5}	Stable
0.00125	0.50	1.03×10^{-4}	7.76×10^{-5}	Stable
0.0015	0.60	Diverged	Diverged	**Unstable**

4.4 Effect of Δy -Explicit Euler

Figure 8 shows the effect of grid refinement on the explicit Euler solution. The ratio $r = 0.4$ is held constant (so Δt decreases proportionally with $(\Delta y)^2$). As N increases from 10 to 80, the numerical solution converges visibly toward the exact solution. The error decreases by approximately a factor of 4 each time Δy is halved, confirming the expected second-order spatial convergence.

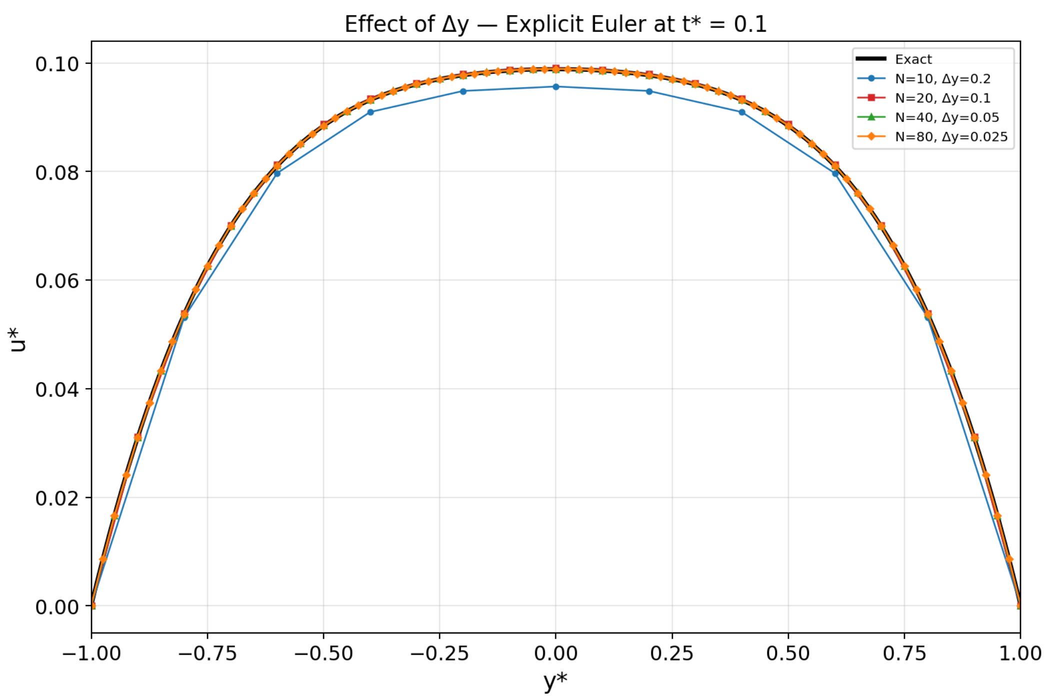

Figure 8: Effect of Δy (grid refinement) on the explicit Euler solution at $t^* = 0.1$ with $r = 0.4$ fixed.

4.5 Effect of r - Explicit Euler

Figure 9 shows velocity profiles for the explicit Euler method at five values of r with $\Delta y = 0.05$ fixed. For $r \leq 0.5$, all solutions are stable and accurate. At $r = 0.6$, the solution exhibits violent oscillations with amplitudes of order 10^3, clearly demonstrating the diffusive instability predicted by the von Neumann analysis (Eq. 31).

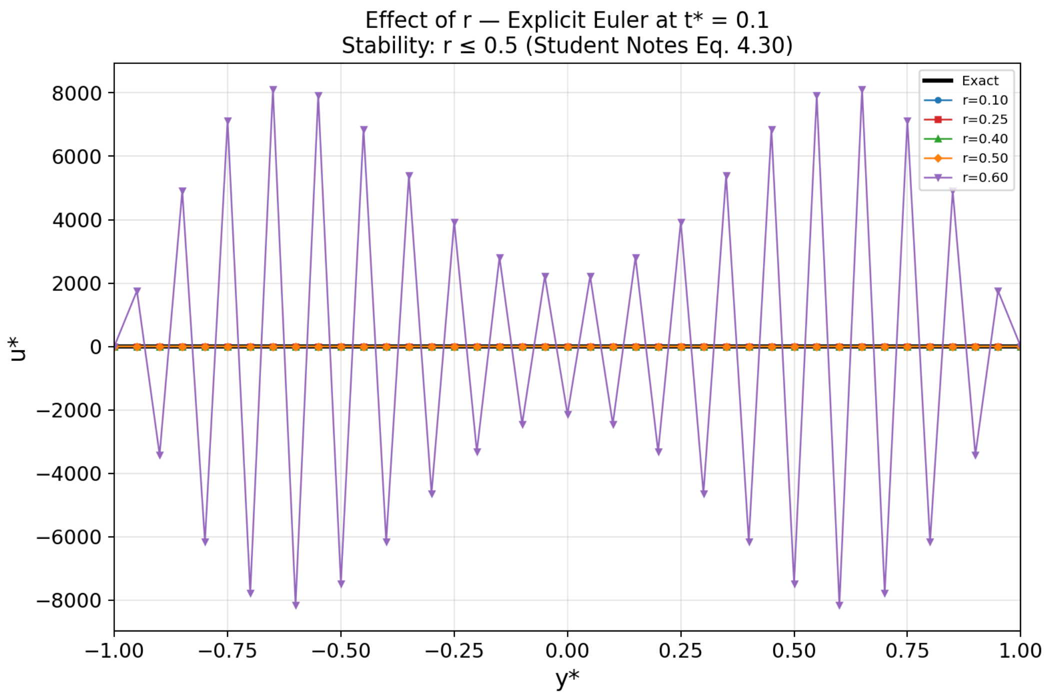

Figure 9: Effect of r on the explicit Euler solution at $t^* = 0.1$ ($N = 40$, $\Delta y = 0.05$). Stability limit: $r \leq 0.5$.

4.6 Effect of Δt - Implicit Euler

Figure 10 presents the implicit Euler results for five values of Δt spanning two orders of magnitude (r from 0.2 to 50). Unlike the explicit method, all cases remain stable, even at $r = 50$, the solution does not blow up. However, the accuracy degrades significantly with increasing Δt: the $r = 50$ case shows a visible deviation from the exact solution. Table 4 confirms that the error grows with Δt, as expected for a first-order method. This demonstrates the key distinction: unconditional stability does not guarantee unconditional accuracy.

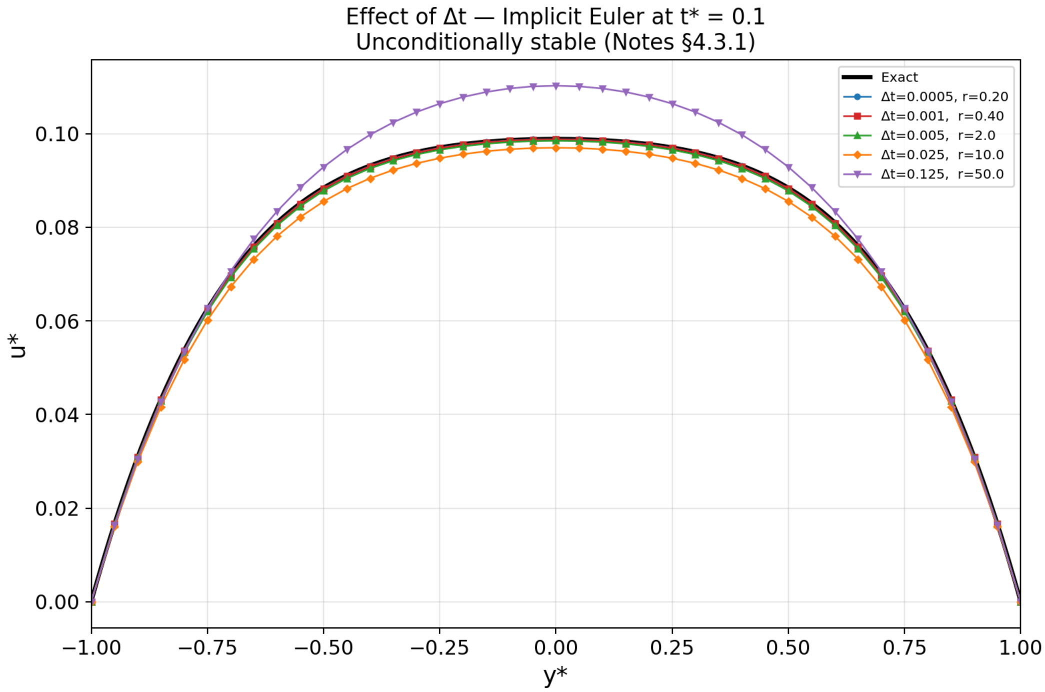

Figure 10: Effect of Δt on the implicit Euler solution at $t^* = 0.1$ ($N = 40$). All cases remain stable regardless of r.

Table 4: Error metrics for the implicit Euler Δt study at $t^* = 0.1$.

Δt	r	E_∞	E_{RMS}
0.0005	0.20	1.12×10^{-4}	8.50×10^{-5}
0.001	0.40	1.74×10^{-4}	1.31×10^{-4}
0.005	2.0	6.58×10^{-4}	4.99×10^{-4}
0.025	10.0	2.95×10^{-3}	2.27×10^{-3}
0.125	50.0	1.33×10^{-2}	1.11×10^{-2}

4.7 Effect of Δy - Implicit Euler

Figure 11 shows the grid convergence for the implicit Euler method at fixed $r = 0.4$. The behavior is similar to the explicit method: the error decreases by approximately a factor of 4 per doubling of N, confirming second-order spatial convergence.

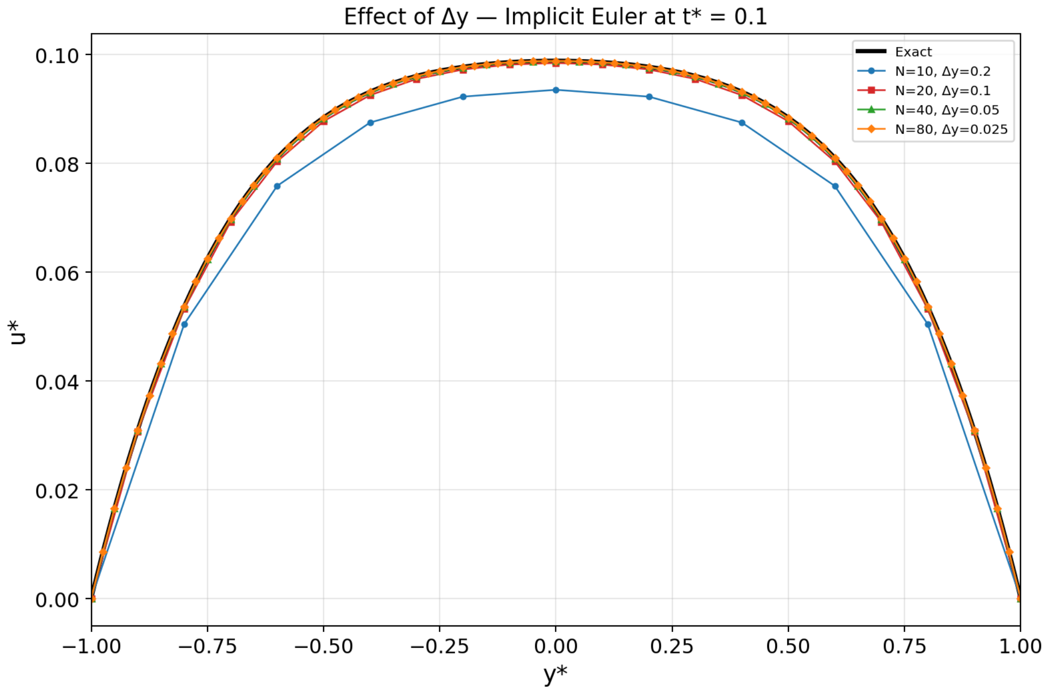

Figure 11: Effect of Δy on the implicit Euler solution at $t^* = 0.1$ with $r = 0.4$ fixed.

4.8 Effect of r - Implicit Euler

Figure 12 demonstrates the unconditional stability of the implicit method across a wide range of r values (0.5 to 50). All cases produce bounded, physically meaningful profiles. At small r, the results are nearly exact. At $r = 50$ (corresponding to $\Delta t = 0.125$, i.e., only 0.8 time steps to reach $t^* = 0.1$), the profile overshoots the exact solution at the centerline but remains qualitatively correct and stable.

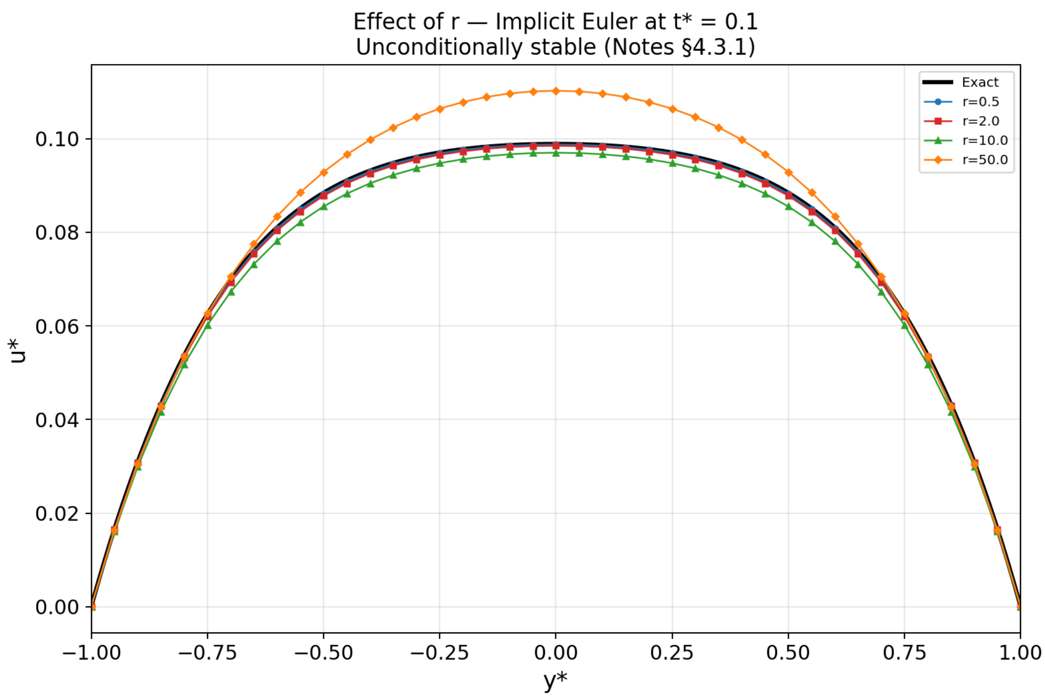

Figure 12: Effect of r on the implicit Euler solution at $t^* = 0.1$ ($N = 40$). The method remains stable for all r values.

4.9 Effect of Δt - Runge-Kutta

Figure 13 shows the RK4 results for four values of Δt. The three stable cases ($r = 0.2, 0.4, 0.6$) all produce nearly identical results with very small errors, reflecting the fourth order temporal accuracy, the time-stepping error is so small that the dominant error contribution comes from the second-order spatial discretization. At $r = 0.8$, the solution diverges violently, indicating that the RK4 stability boundary lies between $r = 0.6$ and $r = 0.8$ for this problem. Table 5 confirms that the stable RK4 errors are nearly constant across Δt values because the spatial error dominates.

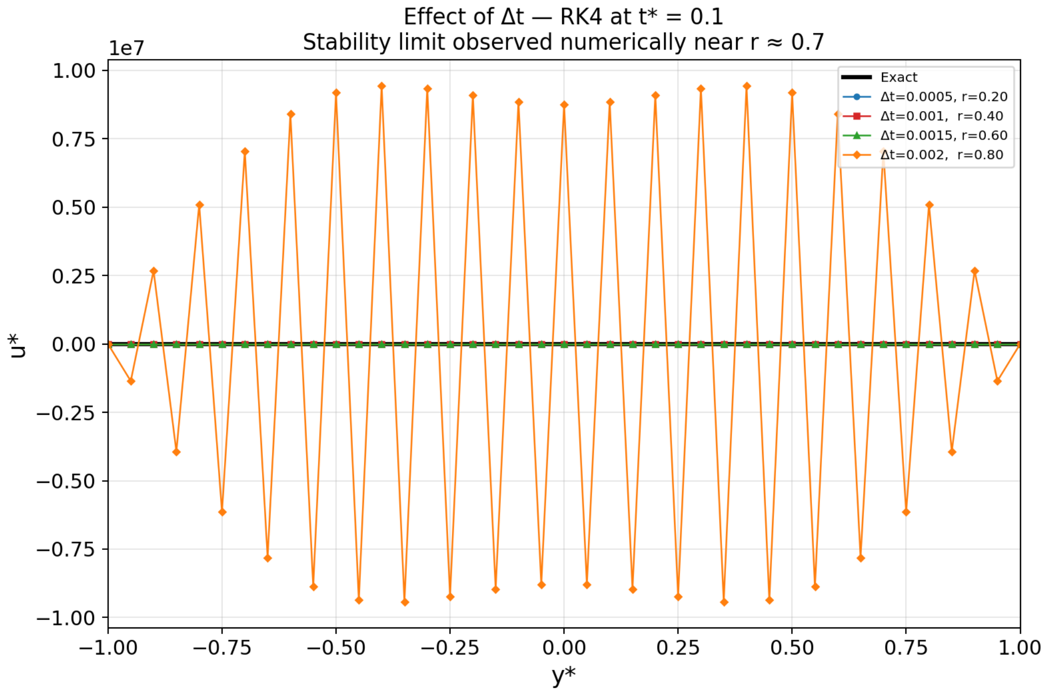

Figure 13: Effect of Δt on the RK4 solution at $t^* = 0.1$ ($N = 40$). The case $r = 0.80$ exceeds the numerically observed stability limit.

Table 5: Error metrics for the RK4 Δt study at $t^* = 0.1$.

Δt	r	E_∞	E_{RMS}	Status
0.0005	0.20	5.11×10^{-5}	3.87×10^{-5}	Stable
0.001	0.40	5.11×10^{-5}	3.87×10^{-5}	Stable
0.0015	0.60	5.11×10^{-5}	3.88×10^{-5}	Stable
0.002	0.80	Diverged	Diverged	**Unstable**

4.10 Effect of Δy - Runge-Kutta

Figure 14 shows the grid convergence for the RK4 method. As with the Euler methods, the error decreases by approximately a factor of 4 per doubling of N, confirming second-order spatial convergence. For the finest grid ($N = 80$), the RK4 error is approximately 1.28×10^{-5}, the smallest among all methods at comparable grid sizes.

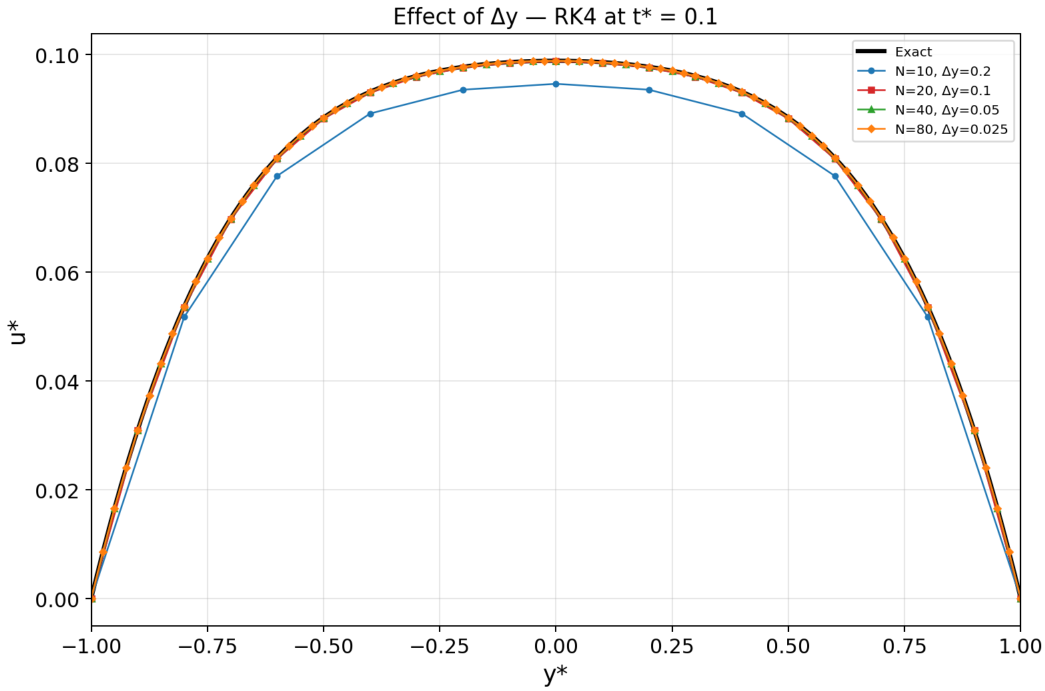

Figure 14: Effect of Δy on the RK4 solution at $t^* = 0.1$ with $r = 0.4$ fixed.

4.11 Effect of r - Runge-Kutta

Figure 15 presents the RK4 solution for five values of r with $\Delta y = 0.05$ fixed. The cases $r = 0.25$ through $r = 0.70$ all produce stable, accurate results. At $r = 0.80$, the solution diverges. This numerically confirms the stability boundary near $r \approx 0.7$. Importantly, unlike the explicit Euler where the stability limit is derived analytically ($r \leq 0.5$), the RK4 limit is an empirical observation specific to this problem.

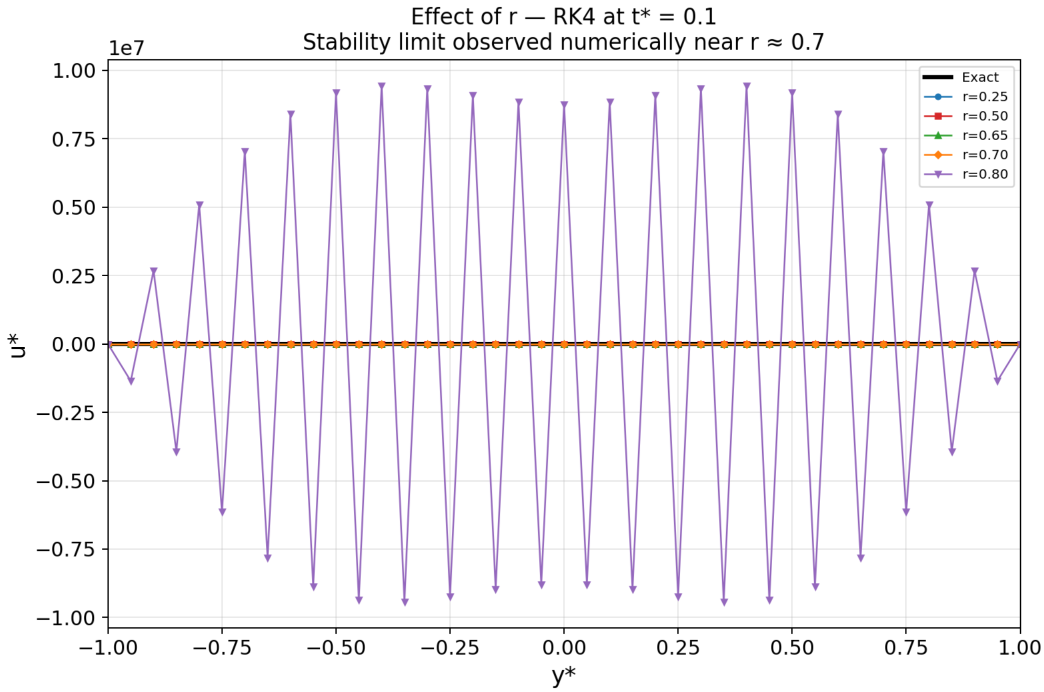

Figure 15: Effect of r on the RK4 solution at $t^* = 0.1$ ($N = 40$). The solution diverges at $r = 0.80$, placing the numerically observed stability limit near $r \approx 0.7$.

4.12 Error Convergence Analysis

Figure 16 presents the E_∞ error at $t^* = 0.5$ as a function of Δt on a log-log scale for all three methods. Dashed reference lines of slope 1 (first order) and slope 4 (fourth order) are included for comparison.

The explicit and implicit Euler methods show approximately first order convergence ($E_\infty \propto \Delta t$), consistent with their $\mathcal{O}(\Delta t)$ truncation error. The RK4 errors remain nearly flat across the tested Δt range, indicating that the spatial discretization error $\mathcal{O}((\Delta y)^2)$ dominates over the temporal error $\mathcal{O}((\Delta t)^4)$. To observe the fourth order temporal convergence of RK4, a finer spatial grid would be required to reduce the spatial error below the temporal error.

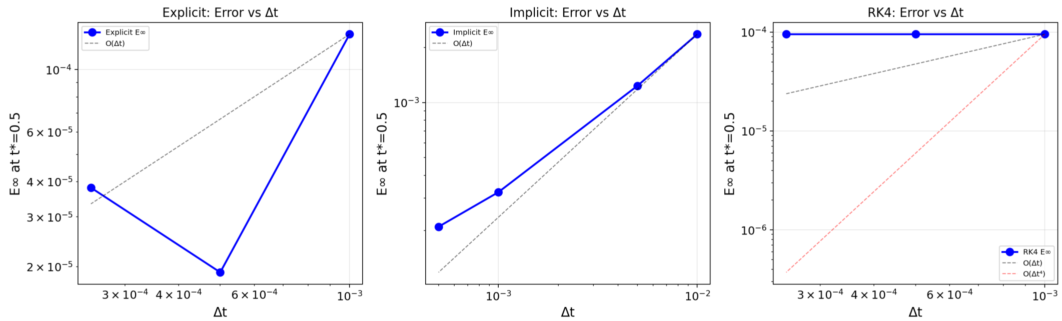

Figure 16: E_∞ error at $t^* = 0.5$ versus Δt for all three methods ($N = 40$). Dashed lines show reference slopes of $\mathcal{O}(\Delta t)$ and $\mathcal{O}((\Delta t)^4)$.

5 Conclusions

This project investigated the time developing, pressure driven flow between two infinite parallel plates. The governing Navier-Stokes equation was simplified to a one dimensional unsteady diffusion equation with a constant source term and nondimensionalized on the domain $-1 \leq y^* \leq 1$, yielding a parameter free formulation. An exact analytical solution was derived using a decomposition into steady state and transient parts, with the transient solved via separation of variables and Fourier series expansion. Three finite difference methods were implemented in Python to solve the nondimensional equation numerically: the explicit (forward) Euler method, the implicit (backward) Euler method, and the fourth order Runge-Kutta (RK4) method, all using the second order central difference scheme for the spatial derivative. The numerical solutions were validated against the exact solution through velocity profiles, volumetric flow rate, and wall shear stress, and systematic parameter studies were conducted to investigate the effects of the time step Δt, spatial grid spacing Δy, and the stability parameter $r = \Delta t/(\Delta y)^2$.

The principal findings of this study are as follows. (1) All three methods produce velocity profiles, flow rates, and wall shear stresses in excellent agreement with the exact solution when stable parameters are used. (2) The explicit Euler method is conditionally stable, requiring $r \leq 0.5$; exceeding this limit produces rapidly growing oscillations that destroy the solution, in exact agreement with the von Neumann stability analysis. (3) The implicit Euler method is unconditionally stable, it produces bounded solutions even at very large values of r (tested up to $r = 50$) but accuracy degrades significantly with increasing Δt, demonstrating that stability does not imply accuracy. (4) The RK4 method provides the highest temporal accuracy among the three methods. For the baseline grid, RK4 errors are dominated by the second order spatial discretization rather than the fourth order temporal discretization. (5) The RK4 stability boundary was numerically observed near $r \approx 0.7$, which is more permissive than the explicit Euler limit of $r = 0.5$ but still imposes a conditional restriction. (6) All three methods exhibit second order spatial convergence: the error decreases by approximately a factor of four each time Δy is halved at fixed r, consistent with the $\mathcal{O}((\Delta y)^2)$ truncation error of the central difference scheme.

Several extensions of this work could be pursued. First, the Crank Nicolson method, which combines the stability advantages of implicit schemes with second order temporal accuracy, could be implemented and compared against the three methods studied here to determine whether it offers a better accuracy-stability trade off. Second, higher order spatial discretizations (such as the fourth order compact scheme) could be employed to reduce the spatial error and allow the fourth order temporal accuracy of RK4 to be fully realized. Third, the present study could be extended to two dimensional configurations, such as the lid driven cavity flow, where the coupled, nonlinear nature of the Navier-Stokes equations introduces additional challenges related to pressure-velocity coupling and iterative solution methods. Finally, adaptive time stepping strategies could be investigated to automatically select the largest stable and accurate Δt at each step, improving computational efficiency without sacrificing solution quality.

References

[1] F. Mashayek, "Computational Fluid Dynamics — AME 431/531 Lecture Notes," Department of Aerospace and Mechanical Engineering, University of Arizona, Spring 2026.

[2] J. H. Ferziger and M. Perić, *Computational Methods for Fluid Dynamics*, 3rd ed. Springer-Verlag, Berlin, 2002.